

Assignment 2: Sanskrit-to-English Neural Machine Translation

1. Introduction

Neural Machine Translation (NMT) is the concept to translate automatically the text from one language to another using deep learning models. Sanskrit is a morphologically rich and low-resource language, making machine translation a challenging task. The objective of this assignment was to develop a Sanskrit-to-English translation system using the provided parallel corpus.

A pretrained **facebook/nllb-200-distilled-600M** multilingual Neural Machine Translation model was fine-tuned on the given Sanskrit-English dataset. The model was trained using the Hugging Face Transformers framework and PyTorch. Beam Search decoding was used during inference to improve translation quality.

2. Model Architecture

The solution uses the **NLLB-200 Distilled 600M** pretrained encoder-decoder Transformer architecture.

Encoder

- Encodes Sanskrit input sentence and makes them as contextual embeddings.
- Multi-head self-attention is for capturing long-range dependencies.
- Positional embeddings preserves the word order.

Decoder

- Generates English translation autoregressively.
- Uses masked selfattention and crossattention over encoder outputs.

Attention Mechanism

The Transformer attention mechanism enables the decoder to focus on relevant Sanskrit words while generating each English token.

Beam Search

Beam Search with **5 beams** was used during inference to generate higher-quality translations compared to greedy decoding.

3. Training Procedure

Dataset

- Training sentences: **10,000**
- Validation sentences: **1,000**

- Test sentences: **1,000**

Preprocessing

- UTF-8 encoded CSV files
- NLLB tokenizer
- Maximum sequence length: **128 tokens**

Hyperparameters

Parameter	Value
Model	facebook/nllb-200-distilled-600M
Epochs	5
Learning Rate	3e-5
Optimizer	AdamW
Batch Size	4
Gradient Accumulation	4
Mixed Precision	FP16
Random Seed	42
Beam Size	5

The best checkpoint was automatically selected based on validation performance.

4. Results

Metric	Score
BLEU Score	29.85
BERTScore F1(rescale_with_baseline=True)	0.5982
Training Loss	1.3461
Validation Loss	1.3867
Epochs	5
Inference Time(in seconds)	584.29

Metric	Score
Model Parameters	615,073,792

Additional Observation:

On using the rescale_with_baseline=False, the BERTScore F1 achieved was 0.9322.

Training Observations

Validation BLEU steadily improved during training.

Epoch Validation BLEU

1	28.00
2	29.38
3	30.54
4	30.30
5	30.69

Training loss consistently decreased while validation loss stabilized after the third epoch, indicating successful convergence without significant overfitting.

5. Translation Examples

Sanskrit	Reference	Model Prediction	Observation
एक्लिप्स इति प्रोग्रामर् कृते दोषान्वेषणे अपि साहाय्यं करोति।	Eclipse also helps the programmer to find out errors.	It also helps in troubleshooting Eclipse program.	Correct semantic meaning with minor wording variation.
तदा, तत्स्वयं ड्राइवर निमित्तम् अन्वेष्यति।	Then it will automatically begin searching for drivers.	Then it will automatically search for the driver.	Accurate translation with slight grammatical variation.
अपरं द्वितीयमुद्रायां...	And when he had opened the second seal...	And when he had opened the second seal...	Preserved overall meaning with minor lexical differences.
विश्वासकारणादेव...	We also believe and therefore speak...	By faith I speak... We also have the spirit of faith...	Semantic meaning retained despite paraphrasing.

Sanskrit	Reference	Model Prediction	Observation
सर्वेभ्यः इटरेशन्...	The iterator will be set...	The iterator is set to the index value...	Technical terminology translated correctly.

6. Experiments & Observations

- Fine-tuning the pretrained NLLB model significantly improved translation quality despite the relatively small training corpus.
- BLEU increased consistently from **28.00** to **30.69** over five epochs.
- The model translated technical and instructional sentences accurately.
- Semantic meaning was generally preserved even when exact wording differed from the reference translation.
- Longer and more complex religious or literary sentences occasionally showed paraphrasing or slight lexical substitutions.
- Beam Search improved fluency and coherence of generated translations.

7. Discussion

The pretrained NLLB model proved effective for Sanskrit-to-English translation. Transfer learning enabled strong performance using only 10,000 parallel sentence pairs. The model generalized well across both technical and general-domain sentences. The primary limitations were occasional paraphrasing and reduced accuracy on long or highly complex Sanskrit sentences.

8. References

1. Costa-jussà et al., *No Language Left Behind: Scaling Human-Centered Machine Translation*, 2022.
2. Papineni et al., *BLEU: A Method for Automatic Evaluation of Machine Translation*, ACL 2002.
3. Zhang et al., *BERTScore: Evaluating Text Generation with BERT*, ICLR 2020.
4. Hugging Face Transformers Documentation.